In addition to the online homework, please hand in the following problems:

Problem 1: Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for each of the following:

a. $\mathbf{F} = [2y, 3z, x]$ and C: consists of the straight paths from $(0,2,0)$ to $(2,2,0)$ to $(2,2,1)$ and back to $(0,2,0)$

b. $\mathbf{F} = [2y, 3z, x]$ and C: consists of the straight paths from $(0,0,0)$ to $(0,2,0)$ to $(1,1,1)$ and back to $(0,0,0)$

c. $\mathbf{F} = y\,\mathbf{i} + (2 - 3yz)\,\mathbf{j} + (x^2 + y^2)\,\mathbf{k}$ and C is the boundary of the side of the cylinder $x^2 + z^2 = 16$ in the first octant with $0 \leq y \leq 2$.